

QA Automation capstone: Project Description

Project Name: - Enterprise E-Commerce UI Automation Framework using Playwright and Java Script.

Project Overview: -

The Enterprise E-Commerce UI Automation Framework using Playwright and JavaScript is designed to validate the most critical workflows of the NIKKA website. It covers authentication, product search, cart, address and shipping, payment and checkout, order history, API services, user profile, and customer support. The framework emphasizes scalability, resiliency, and maintainability, ensuring seamless execution across Chromium, Firefox, and WebKit. By integrating UI automation, API validation, visual regression, and AI-assisted test generation, it delivers enterprise-grade reliability, performance, and reporting suitable for production environments.

Website name: - NIKKA

Website URL: -<https://www.nykaa.com/>

Planned Services: -

1. Authentication & User Accounts
2. Product Search & Discovery
3. Cart Management
4. Address & Shipping
5. Payment & Checkout
6. Order History & Tracking
7. API Service (Mocking & Backend Validation)
8. User Profile Management

Test Case Distribution

- **Services:** 8
- **Test cases per service:** ~15
- **Total per browser:** 120
- **Browsers:** Chromium, Firefox, WebKit
- **Grand Total Executions:** 360

Conclusion

This capstone project establishes a future-ready automation solution that balances technical innovation with business impact. By validating end-to-end customer journeys and integrating AI, API resilience, and enterprise reporting, it enhances trust, usability, and operational stability. The framework positions NIKKA as a benchmark in modern quality engineering, showcasing best practices in scalable automation and intelligent test design.